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VERIFICATION OF MAXIMUM POWER TRANSFER THEOREM

AIM:

To measure the power absorbed in a load and to verify that the power absorbed in a load is maximum only when load resistance is equal to the source resistance.

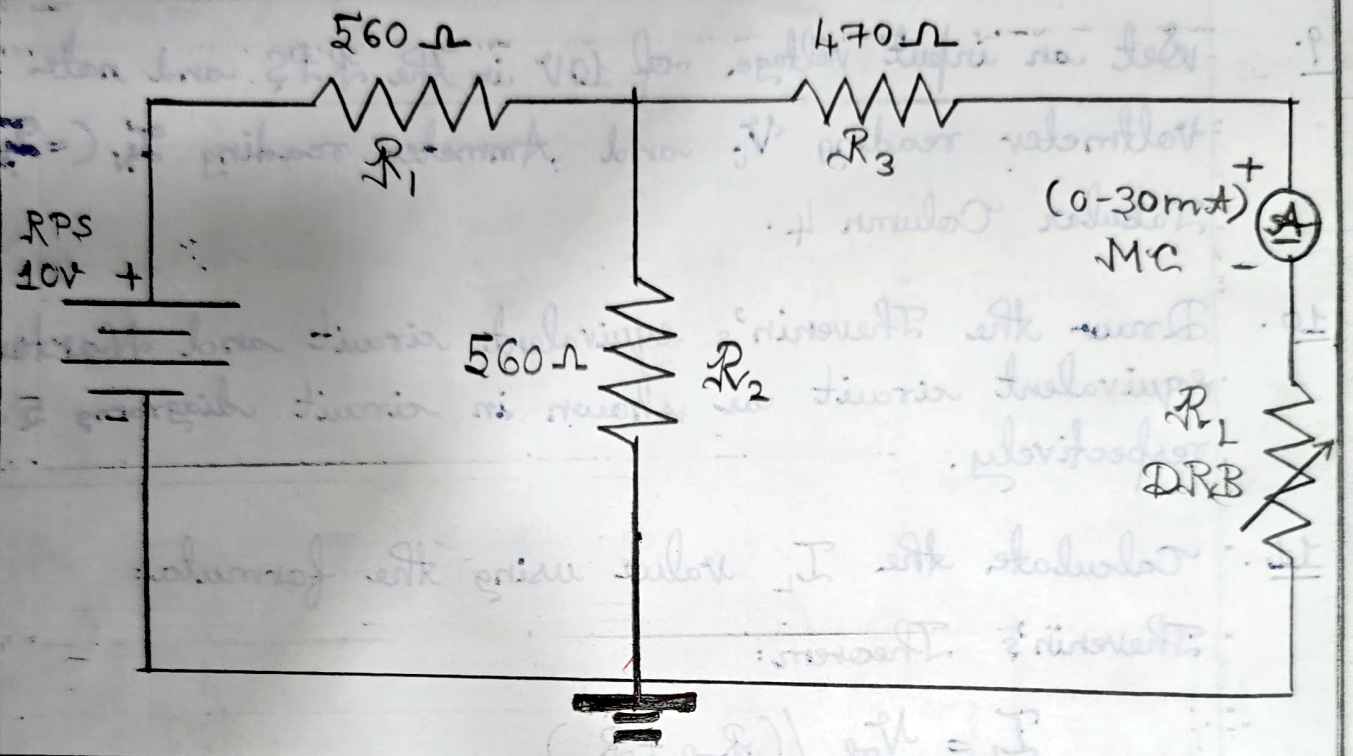
APPARATUS REQUIRED:

SL.No.	APPARATUS	SPECIFICATIONS	QUANTITY
1.	Voltmeter	(0-15)V M.C.	1
2.	Ammeter	(0-500mA) M.C.	1
3.	Resistors	560Ω, 470Ω	2, 1 (respectively)
4.	RPS (DC Supply)	15V	1

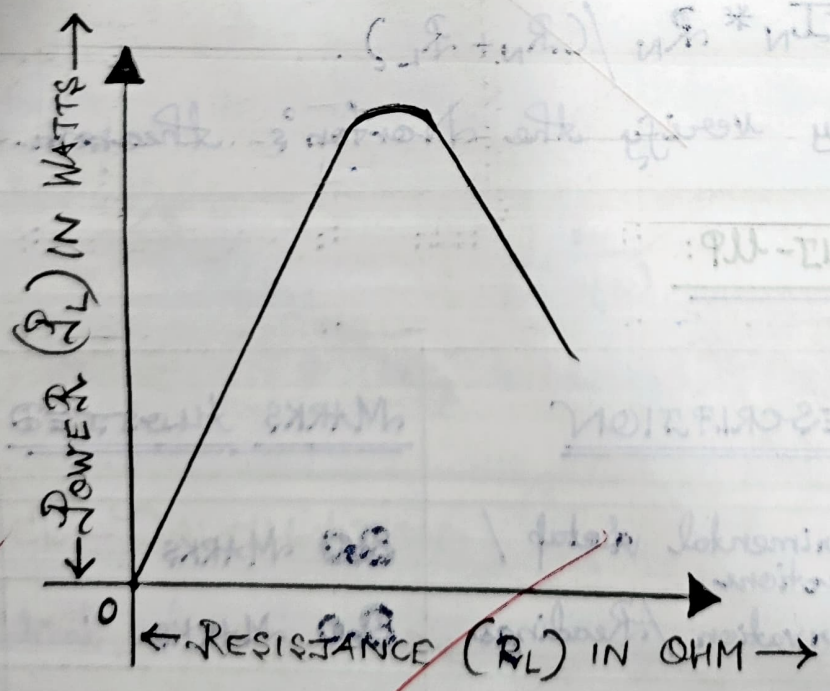
PROCEDURE:

1. Make connection as per circuit diagrams given.
2. Change the resistors R_L whose value close to R_{Th}, measure the corresponding V_L, I_L and calculate the power P_L and enter into the table (2).
3. Plot a graph between R_L and P_L, and find the R_L corresponding to the maximum power transfer.
4. Verify the measured values of R_L at maximum power transfer as same as calculated and found graphically.

CIRCUIT DIAGRAM:



MODEL GRAPH:



⊛. TABULAR COLUMNS:

(1). FOR THEORETICAL CALCULATION : $R_{TH} = 750\,\Omega$.

S.No.	LOAD RESISTANCE (R_L) IN OHM	LOAD CURRENT (I_L) IN mA	LOAD VOLTAGE (V_L) IN VOLTS	LOAD POWER (P_L) IN WATTS
1.	500	4.00	2.00	0.0080
2.	750	3.33	2.50	0.00833
3.	1000	2.80	2.35	0.00816
4.	1500	2.00	3.00	0.00600
5.	2000	1.67	3.33	0.00566

(2). FOR PRACTICAL CALCULATION:

S.No.	LOAD RESISTANCE (R_L) IN OHM	LOAD CURRENT (I_L) IN AMPS	LOAD VOLTAGE (V_L) IN VOLTS	LOAD POWER (P_L) IN WATTS
1.	500	3.90	1.95	0.0076
2.	750	3.25	2.44	0.0792
3.	1000	2.80	2.80	0.00784
4.	1500	1.95	2.93	0.00571
5.	2000	1.60	3.20	0.00512

O/P voltage 24 V

⊛. MODEL CALCULATION:

Given :-

→ $R1 = 560\,\Omega$.

→ $R2 = 560\,\Omega$.

→ $R3 = 470\,\Omega$.

→ Supply Voltage = 10V.

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MARKS SPLIT-UP:

SL.NO.	DESCRIPTION	MARKS ALLOTTED	MARKS OBTAINED
1.	Experimental Setup/ Connections	20 MARKS	20
2.	Observation/Readings	20 MARKS	20
3.	Calculations/Results	20 MARKS	20
4.	Viva	20 MARKS	20
5.	Record	20 MARKS	20
		TOTAL:	100

RESULT:

Thus, the maximum power transfer theorem is verified - practically and theoretically.

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CALCULATION:

Step 1:- Find Rth (R_{th}):

$$\Rightarrow R_{th} = R_3 + \left(\frac{1}{R_1} + \frac{1}{R_2}\right)$$

$$\Rightarrow R_{th} = 470 + \left(\frac{560 \times 560}{560 + 560}\right)$$

$$\Rightarrow R_{th} = (470 + 280)\,\Omega$$

$$= 750\,\Omega.$$

Step 2:- Find V_{th}:

$$\Rightarrow V_{th} = 10 \times \frac{R_2}{R1 + R2}$$

$$= 10 \times \frac{560}{560 + 560}$$

$$= (10 \times 0.5)\,V$$

$$= 5\,V.$$

Step 3:- For $R_L = 750\,\Omega$:

$\Rightarrow R_L = R_{th}$

$\Rightarrow R_L = 750\,\Omega.$

$$\Rightarrow I_L = \frac{V_{th}}{R_{th} + R_L}$$

$$= \frac{5}{760 + 760}$$

$$\approx 3.33\,mA$$

$$\approx 0.00333\,A.$$

$$\Rightarrow V_L = I_L \times R_L$$

$$= 0.0033 \times 750$$

$$= 2.5\,V.$$

$$\Rightarrow P_L = V_L \times I_L$$

$$= 2.5 \times 0.00333$$

$$\approx 8.33\,mW.$$

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RESULT:

Thus, the maximum power transfer theorem is verified practically and theoretically.